

MTH 166

Lecture-33

Line Integral

Unit 6: Vector Calculus-II

(Book: Advanced Engineering Mathematics by Jain and Iyengar, Chapter-15)

Topic:

Vector Integral Calculus

Learning Outcomes:

1. To calculate Line integral w.r.t to an arc length
2. To calculate Line integral of a vector field (Work Done)

Line Integral w.r.t an Arc Length

Let the parametric representation of curve C be:

$$C: x = x(t), y = y(t), z = z(t); \quad a \leq t \leq b.$$

Let $f(x, y, z)$ be a scalar field function.

Then line integral of scalar function f over curve C w.r.t arc length s is given by:

$$\begin{aligned} I &= \int_C f(x, y, z) ds \\ &= \int_{t=a}^{t=b} \left[f(x(t), y(t), z(t)) \frac{ds}{dt} \right] dt \\ &= \int_{t=a}^{t=b} \left[f(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \right] dt \end{aligned}$$

Problem 1. Evaluate $\int_C f(x, y, z)ds$ where $f(x, y, z) = 2x + 3y$ and curve C is:

$$C: x = t, y = 2t, z = 3t, 0 \leq t \leq 3.$$

Solution. $C: x = t, y = 2t, z = 3t$

$$\Rightarrow \frac{dx}{dt} = 1, \quad \frac{dy}{dt} = 2, \quad \frac{dz}{dt} = 3$$

$$f(x, y, z) = 2x + 3y \quad \Rightarrow f(t) = 2(t) + 3(2t) = 8t$$

Then line integral of scalar function f over curve C w.r.t arc length s is given by:

$$I = \int_C f(x, y, z)ds = \int_{t=a}^{t=b} \left[f(x(t), y(t), z(t)) \frac{ds}{dt} \right] dt$$

$$= \int_{t=a}^{t=b} \left[f(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \right] dt$$

$$I = \int_{t=0}^{t=3} \left[8t\sqrt{(1)^2 + (2)^2 + (3)^2} \right] dt$$

$$= 8\sqrt{14} \int_{t=0}^{t=3} t dt$$

$$= 8\sqrt{14} \left[\frac{t^2}{2} \right]_0^3$$

$$= 8\sqrt{14} \left[\frac{9}{2} - 0 \right]$$

$$= 36\sqrt{14} \quad \mathbf{Answer.}$$

Problem 2. Evaluate $\int_C f(x, y)ds$ where $f(x, y) = x^2y$ and curve C is:

$$C: x = 3 \cos t, y = 3 \sin t, 0 \leq t \leq \frac{\pi}{2}.$$

Solution. $C: x = 3 \cos t, y = 3 \sin t$

$$\Rightarrow \frac{dx}{dt} = -3 \sin t, \quad \frac{dy}{dt} = 3 \cos t$$

$$f(x, y) = x^2y \Rightarrow f(t) = (3 \cos t)^2(3 \sin t) = 27 \cos^2 t \sin t$$

Then line integral of scalar function f over curve C w.r.t arc length s is given by:

$$I = \int_C f(x, y)ds = \int_{t=a}^{t=b} \left[f(x(t), y(t)) \frac{ds}{dt} \right] dt$$

$$= \int_{t=a}^{t=b} \left[f(t) \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} \right] dt$$

$$I = \int_{t=0}^{t=\frac{\pi}{2}} 27 \cos^2 t \sin t \left[\sqrt{(-3 \sin t)^2 + (3 \cos t)^2} \right] dt$$

$$= 27 \times 3 \int_{t=0}^{t=\frac{\pi}{2}} \cos^2 t \sin t dt$$

$$= -81 \int_{t=0}^{t=\frac{\pi}{2}} \cos^2 t (-\sin t) dt$$

$$= -81 \left[\frac{(\cos t)^3}{3} \right]_0^{\frac{\pi}{2}}$$

$$= -27 \left[\left(\cos \frac{\pi}{2} \right)^3 - (\cos 0)^3 \right]$$

$$= -27(0 - 1) = 27 \text{ Answer.}$$

Polling quiz

The line integral $\int_C f(x, y, z) ds$ of function $f(x, y, z)$ which is continuous over the simple smooth curve $C(x(t), y(t), z(t))$, $a \leq t \leq b$, with respect to the arc length s is given by:

(a) $\int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$

(b) $\int_a^b f(x(t), y(t), z(t)) dt$

(c) $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$

(d) None of these.

Line Integral of a Vector Field (Work Done)

Let us consider a vector field $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

We write $\overrightarrow{v(t)} = v_1(t)\hat{i} + v_2(t)\hat{j} + v_3(t)\hat{k}; \quad a \leq t \leq b$

Let curve C is represented by: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

We write $C: \overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$

Then line integral of vector $\vec{v}$ over curve C given by:

$$\begin{aligned} I &= \int_C \vec{v} \cdot d\vec{r} = \int_C \left[\overrightarrow{v(t)} \cdot \frac{d\vec{r}}{dt} \right] dt \\ &= \int_{t=a}^{t=b} \left[(v_1(t)\hat{i} + v_2(t)\hat{j} + v_3(t)\hat{k}) \cdot \left(\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} \right) \right] dt \\ &= \int_{t=a}^{t=b} \left[\left(v_1(t) \frac{dx}{dt} + v_2(t) \frac{dy}{dt} + v_3(t) \frac{dz}{dt} \right) \right] dt \quad \text{This also called as **Work Done**.} \end{aligned}$$

Problem 1. Evaluate $\int_C \vec{v} \cdot \overrightarrow{dr}$ where $\vec{v} = xy\hat{i} + y^2\hat{j} + e^z\hat{k}$ and curve c is given by:

$$C: x = t^2, y = 2t, z = t, 0 \leq t \leq 1.$$

Solution. We write $C: \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \Rightarrow \overrightarrow{r(t)} = t^2\hat{i} + 2t\hat{j} + t\hat{k}$

$$\Rightarrow \frac{dx}{dt} = 2t, \frac{dy}{dt} = 2, \frac{dz}{dt} = 1$$

$$\vec{v} = xy\hat{i} + y^2\hat{j} + e^z\hat{k} \Rightarrow \overrightarrow{v(t)} = 2t^3\hat{i} + 4t^2\hat{j} + e^t\hat{k}$$

Then line integral of vector $\vec{v}$ over curve C given by:

$$I = \int_C \vec{v} \cdot \overrightarrow{dr} = \int_C \left[\overrightarrow{v(t)} \cdot \frac{\overrightarrow{dr}}{dt} \right] dt$$

$$= \int_{t=a}^{t=b} \left[\left(v_1(t) \frac{dx}{dt} + v_2(t) \frac{dy}{dt} + v_3(t) \frac{dz}{dt} \right) \right] dt$$

$$\begin{aligned}
I &= \int_{t=a}^{t=b} \left[\left(v_1(t) \frac{dx}{dt} + v_2(t) \frac{dy}{dt} + v_3(t) \frac{dz}{dt} \right) \right] dt \\
&= \int_{t=0}^{t=1} \left[\left((2t^3)(2t) + (4t^2)(2) + (e^t)(1) \right) \right] dt \\
&= \int_{t=0}^{t=1} [4t^4 + 8t^2 + e^t] dt \\
&= \left[\frac{4}{5} t^5 + \frac{8}{3} t^3 + e^t \right]_{t=0}^{t=1} \\
&= \left[\frac{4}{5} + \frac{8}{3} + e \right] - [0 + 0 + 1] \\
&= \frac{37}{15} + e \quad \textbf{Answer.}
\end{aligned}$$

Problem 2. Evaluate $\int_C \vec{v} \cdot \overrightarrow{dr}$ where $\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$ and C is the line segment from $(1,2,2)$ to $(3,6,6)$. **Or.**

Find work done by force $\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$ in moving a particle from $(1,2,2)$ to $(3,6,6)$.

Solution. Two point form of line C is given by:

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1} = t \quad (\text{Say})$$

$$\Rightarrow \frac{x-1}{3-1} = \frac{y-2}{6-2} = \frac{z-2}{6-2} = t$$

$$\Rightarrow \frac{x-1}{2} = \frac{y-2}{4} = \frac{z-2}{4} = t$$

$$\Rightarrow x = 2t + 1, \quad y = 4t + 2, \quad z = 4t + 2, \quad (\text{when } x = 1, t = 0 \text{ and } x = 3, t = 1)$$

$$\text{So, } C: \quad \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad \Rightarrow \overrightarrow{r(t)} = (2t + 1)\hat{i} + (4t + 2)\hat{j} + (4t + 2)\hat{k}$$

$$\Rightarrow \frac{dx}{dt} = 2, \quad \frac{dy}{dt} = 4, \quad \frac{dz}{dt} = 4$$

$$\vec{v} = x\hat{i} + y\hat{j} + z\hat{k} \Rightarrow \overrightarrow{r(t)} = (2t + 1)\hat{i} + (4t + 2)\hat{j} + (4t + 2)\hat{k}$$

Then line integral of vector $\vec{v}$ over curve C given by:

$$I = \int_C \vec{v} \cdot \overrightarrow{dr} = \int_C \left[\overrightarrow{v(t)} \cdot \frac{\overrightarrow{dr}}{dt} \right] dt$$

$$I = \int_{t=a}^{t=b} \left[\left(v_1(t) \frac{dx}{dt} + v_2(t) \frac{dy}{dt} + v_3(t) \frac{dz}{dt} \right) \right] dt$$

$$= \int_{t=0}^{t=1} \left[((2t + 1)(2) + (4t + 2)(4) + (4t + 2)(4)) \right] dt$$

$$= \int_{t=0}^{t=1} [36t + 18] dt = \left[36 \frac{t^2}{2} + 18t \right]_{t=0}^{t=1} = (18 + 18) - 0 = 36 \text{ **Answer.**}$$

